Module 1
Advanced Retrievers for RAG

Welcome to the Course

- Course Introduction
  Video • 4 min
- Course Overview
  Reading • 5 min
- RAG and Agentic AI Professional Certificate Overview
  Video • 5 min
- Reading: Helpful Tips for Course Completion
  Ungraded Plugin • 5 min

Work with Advanced Retrievers in LangChain

- Explore Advanced Retrievers in Langchain: Part 1
  Video • 2 min
- Explore Advanced Retrievers in Langchain - Part 2
  Video • 5 min
- Lab: Build a Smarter Search with LangChain Context Retrieval
  Ungraded App Item • 1h
- [Optional] Interactive LangChain Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- Practice Quiz: Work with Advanced Retrievers in LangChain
  Practice Assignment • Grade: 85.71%

Work with Advanced Retrievers in LlamaIndex

- Advanced Retrievers in LlamaIndex
  Video • 8 min
- Lab: Explore Advanced Retrievers in LlamaIndex
  Ungraded App Item • 1h
- [Optional] Interactive LlamaIndex Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- Practice Quiz: Work with Advanced Retrievers in LlamaIndex
  Practice Assignment • Grade: 100%

Module Summary and Evaluation

- Summary and Highlights: Advanced Retrievers for RAG
  Reading • 3 min
- Reading: Cheat Sheet: Advanced Retrievers for RAG
  Ungraded Plugin • 15 min
- Graded Quiz: Advanced Retrievers for RAG
  Graded Assignment • Grade: 100%
- [Optional] Meet and Greet
  Discussion Prompt • 2 min

Module 2
Build a Comprehensive RAG Application

Introduction to FAISS for RAG

- Introduction to FAISS and How It compares to ChromaDB
  Video • 5 min
- Lab: Semantic Similarity with FAISS
  Ungraded App Item • 1h
- Reading: Hierarchical Navigable Small World (HNSW)
  Ungraded Plugin • 30 min
- Lab: AI-Powered YouTube Summarizer, QA Tool with RAG, LangChain, FAISS
  Ungraded App Item • 1h
- [Optional] Interactive FAISS/HNSW Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- Practice Quiz: Introduction to FAISS for RAG
  Practice Assignment • Grade: 100%

Module Summary and Evaluation

- Summary and Highlights: Build a Comprehensive RAG Application
  Reading • 2 min
- Cheat Sheet: Build a Comprehensive RAG Application
  Ungraded Plugin • 15 min
- Graded Quiz: Build a Comprehensive RAG Application
  Graded Assignment • Grade: 100%

Course Wrap-Up

- Course Wrap-Up
  Video • 6 min
- **Congratulations and Next Steps**
  Reading • 2 min
- Team and Acknowledgements
  Reading • 2 min

Congratulations and Next Steps

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Congratulations on completing this intermediate course on Advanced RAG with Vector Databases and Retrievers! We hope you enjoyed it and find great satisfaction using your new skills in the workplace or elsewhere.

This course is part of the IBM RAG and Agentic AI Professional Certificate. By completing the program, you've understood how to differentiate between various retrieval patterns and assess their effectiveness in RAG applications. You can implement advanced retrievers and FAISS to optimize information retrieval and similarity search. You can evaluate retrieval strategies and refine AI-driven search capabilities for improved performance. Finally, you can design a comprehensive RAG application by integrating LangChain, FAISS, and a front-end UI using Gradio.

Also, consider this individual course that is part of the professional certificate.

- Vector Databases for RAG: An Introduction

We also encourage you to leave your feedback and rate this course so that we can continue to improve our content.

Good luck!

Mark as completed

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